ROBUSTNESS OF ML PHISHING DETECTORS AGAINST AI-GENERATED LURES

Protecting Your Digital Assets



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WHY THIS MATTERS (THE PROBLEM)

- Phishing targets people, so it stays a top threat.
- AI writes fluent, personalized lures at scale.
- Old detectors may miss them.

The research gap

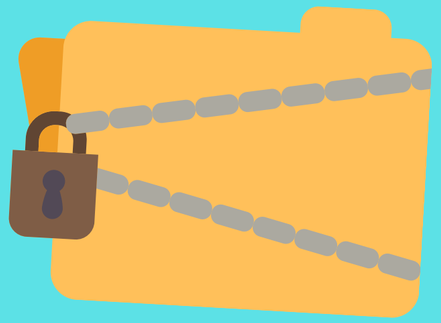
- Most published detectors are evaluated on old datasets and report very high accuracy.
- Few studies test them on LLM-written phishing, or across datasets.
- Gap: we don’t know how much performance drops, which features break, or how to fix it.

4. RESEARCH QUESTION AND HYPOTHESES











RQ1: How much does detector performance drop on AI-generated phishing compared with human-written phishing?

RQ2: Which feature types (text, URL, header) are most robust?

RQ3: Does training on a mix of AI-generated and traditional samples improve robustness?

Hypothesis: text-only models degrade the most, and header/URL features are more robust.

Related work

- Classic ML phishing detection (TF-IDF with SVM, Random Forest).
- Transformer-based detectors.
- Studies on AI-generated phishing.
- Email authentication (SPF/DKIM/DMARC).

6. Proposed approach (method)

- A pipeline diagram: Data collection → Preprocessing → Feature extraction → Model training → Robustness testing → Analysis
- Models: Logistic Regression, Random Forest, and a fine-tuned small transformer (for example DistilBERT).
- Features: text (TF-IDF, embeddings), URL features, header features (sender/reply-to mismatch, SPF/DKIM results).
- Attack set: phishing emails generated with an open-source LLM in a controlled lab setting.

Data

- Human-written phishing: Nazario corpus and public Kaggle phishing email datasets.
- Legitimate email: Enron and SpamAssassin ham.
- AI-generated phishing: generated by you, labeled and stored.
- Mention the known issue of class imbalance and dataset age.

Evaluation plan

Metrics: precision, recall, F1, ROC-AUC, with extra attention to recall on AI-generated emails.

Experiments: (a) in-dataset test, (b) cross-dataset test, (c) human-written vs AI-generated test, (d) feature ablation, (e) retraining with mixed data.

Optional: SHAP or similar to explain what the models rely on.

Option 1: Expected outcome

- Expected outcome: a measured performance gap on AI-written phishing, the most robust feature types, and a way to reduce the gap.

Option 2: Why it matters

- Expected outcome: a measured performance gap on AI-written phishing, the most robust feature types, and a way to reduce the gap.

Option 3: Scope and contribution

- **Scope: email phishing detection with classical ML and a small transformer model.**
- **Contribution: a robustness comparison across text, URL, and header features.**
- **Outcome: evidence-based guidance for building detectors that handle AI-written lures.**

THANK YOU FOR YOUR ATTENTION

